

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative modeler with experience in stochastic processes, approximate likelihoods, sequential testing, and large-scale risk modeling. Builds scalable Python/PyTorch systems and model diagnostics for dependent data on AWS and HPC.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ
M.S. in Statistics, University of Minnesota, Minneapolis, MN

Expected Dec 2026 | GPA: 3.8/4.0
Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, PyTorch, LightGBM, scikit-learn), R, SQL/MySQL, Linux/HPC, AWS EC2

Quantitative Methods: Stochastic processes, MLE, Gaussian processes, approximate likelihood, spectral methods, sequential testing, model validation, GLM/GBM

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling Summer 2026

- Designed a sequential anomaly-detection framework for binary risk indicators by representing cumulative evidence as a random walk and using Brownian-motion approximations to quantify stopping-time uncertainty and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team Summer 2025

- Developed an end-to-end LightGBM pricing pipeline for mid-market property risk - data processing, GLM benchmarking, validation, and diagnostics - across 2.46M+ records on AWS EC2.
- Translated model diagnostics and feature-attribution patterns into pricing and risk-segmentation recommendations for business stakeholders.

QUANTITATIVE RESEARCH AND PROJECTS

Ph.D. Thesis - Scalable Inference and Model Diagnostics for GEMS Ozone Data Sep 2023 - Present

- Developed an advection-aware Vecchia likelihood for nonseparable spatio-temporal Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit 145,008 observations per day.
- Built a leverage-adjusted conditional-eigen diagnostic that localizes covariance misspecification by scale; controlled simulations kept the correct model inside a 95% Brownian-bridge band while rejecting wrong smoothness and nugget specifications.
- Derived missing-data spectral-distortion and expected-periodogram diagnostics; validated the cross-periodogram implementation over 1,000 simulations (MAD 0.00648) and used directional cross-variograms to test advection and wind-direction consistency.

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants Apr 2025

- Built a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance evaluation.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)